Building an AI University

The University of Florida Artificial Intelligence Initiative touches every corner of the university, aiming to reach every corner of the world.

Embracing AI on Every Level

When the University of Florida partnered with NVIDIA to serve as home to one of the world's fastest supercomputers in higher education, it stepped onto the forefront of AI in higher education. Rather than reserve this powerful resource for researchers and faculty in designated colleges or STEM disciplines, UF has used this technology to transform the university as a whole.

Preparing Everyone for the Promise of AI

The World Economic Forum has predicted that 97 million new jobs will require some level of AI talent this year.

All academic disciplines and industries engage with massive amounts of data, and data is at the heart of nearly every facet of modern life. UF has embarked on a university-wide effort to benefit the state and nation, and prepare the 21st century workforce, through producing graduates—and conducting research—that uses AI to transform data into action.

AI Across the Curriculum

Since 2020, UF has embarked on a university-wide effort to produce an AI-enabled workforce for Florida and beyond.
Creating AI-Enabled Workers

AI Ethics

The ethical issues surrounding AI—privacy, transparency, bias, academic integrity—are serious. UF requires ethics courses in its certificate and micro-credential programs.

More About AI Ethics

AI Initiatives

In 2023, UF distributed nearly $19 million to fund AI efforts in 10 colleges. We are investing those funds in a transformational approach to integrating AI with all curriculum and interdisciplinary research.
Our Strategic Investment in AI

A Brief Timeline

Since 2020, UF has made fast leaps to becoming an AI university.

Pre-2020

2012

- UF builds a 1.6 MW data center

2013

- UF Information Technology (UFIT) purchases the 1st-generation HiPerGator, which enters production with 16,000 AMD CPU cores; older clusters with 8,000 CPU cores are deprecated.
- UF develops a sustainability model for high-performance computing (HPC) by enabling faculty, departments, institutes, and colleges to purchase CPU and storage access using startup funds and grants. The university carries staff, data center, and energy costs.

2015

- **UF policy:** Faculty startup funds may be used only to purchase access to HiPerGator; distributed clusters are no longer permitted.
- UF purchases a 2nd-generation HiPerGator system to meet demand, adding 30,000 Intel CPU cores and 40 NVIDIA K80 GPUs.

2017

- **UF Policy:** All research contracts involving restricted data must use the secure enclave in HiPerGator, which meets NIST 800-171 and NIST 800-53 Moderate compliance requirements.
- **UF Policy:** Data use agreements (DUAs) may be signed only by the Office of Research.

2019

- UF adds 560 NVIDIA 2080 Ti GPUs and 48 RTX 6000 GPUs to meet growing demand for AI workloads and advanced simulation and modeling.

INITIAL VISION

- When Chris Malachowsky, a UF alum and co-founder of NVIDIA, asked what UF would do with a supercomputer, Provost Joe Glover responded, “We would build an AI university and integrate AI across the curriculum.”
- UF’s accountability plans for 2020 and 2021 highlight the intention to deepen engagement in AI.
- University faculty and departments already using AI began collaborating on AI across the curriculum.

2020

Building the Foundation

- UF invests $15M to upgrade the datacenter's power and cooling capacity from 1.6 MW to 3.2 MW.
- HiPerGator's 3rd generation, with 40,000 AMD CPU cores, was acquired using funds generated by the sustainable operation of HiPerGator.
- Chris Malachowsky and NVIDIA are donating the $50M NVIDIA DGX A100 SuperPOD as part of HiPerGator's 3rd generation, featuring 1,120 NVIDIA Ampere A100 GPUs.
- Provost Glover works with the deans of all 16 colleges to establish an AI working group in each college to guide course development and to develop a research strategy tailored to each college's specific needs and aligned with the "AI Across the Curriculum" vision.

2021

AI Faculty Hiring Initiative

- UF launches a two-year AI hiring initiative, bringing on 106 new AI faculty members across colleges.

Build Infrastructure and Support Services

- HiPerGator 3rd generation goes into production; HiPerGator 1st generation is retired.
- UFIT hires training and support staff to support faculty, students, and staff, and develops extensive AI training materials, sessions, and workshops, leveraging NVIDIA Deep Learning Institute (DLI) materials.

2022

UF Launches AI Across the Curriculum

- UF launches the Artificial Intelligence Academic Initiative (AI2) Center, a central hub for all AI academic efforts at UF. AI Days and AI Scholars initiatives launch under the AI² Center.

Build Infrastructure

- The AI Initiative generates substantial data. The Orange and Blue storage systems of HiPerGator have been expanded from 3 PB to 16 PB (Orange) and from 4 PB to 7.2 PB (Blue), using funds from the sustainable operation of HiPerGator.

2023

- The State of Florida made a strategic investment in artificial intelligence. UF allocated $18.8 million of that investment to fund 15 projects across 10 colleges.
- Following the November 2022 announcement of ChatGPT, universities from across the U.S. and the world visited UF to learn how to operate an AI university.

2024

- **Accreditation:** Certification program launches: All students can learn about AI, regardless of their major. The program covers:
    - the fundamentals of AI,
    - AI ethics, and
    - A third applied AI course focuses on their field of study.

- An AI career expert is hired to work between the AI² Center and the Career Connections Center.
- AI class designation and tagging continue.
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- NaviGator AI launches to provide all faculty, students, and staff a flexible, extensible, usage-logging service for using over 30 large Language models (LLM). NaviGator AI provides access to ChatGPT in Azure, Claude in AWS, Gemini in GCP, and numerous open-source models running on HiPerGator.
- **UF policy:** All faculty, staff, and students must use NaviGator AI for university business.

2025

Growth and Updated Technology Continues

- HiPerGator 4th generation, with 20,000 AMD CPU cores and a replacement for Blue storage, acquired 17 PB with funds from the sustainable operation of HiPerGator, increasing compute power by a factor of 7.
- Planned investment of $24 million to upgrade the donated NVIDIA DGX A100 SuperPOD with the latest technology, the NVIDIA DGX B200 SuperPOD from 2020, with 504 Blackwell 200 GPUs.

AI Curriculum Strategy

- "AI Across the Curriculum: Governance, Taxonomy, and Institutional Strategy" by Brittany Wise.

2026

- UF buys 16 DGX B300 GPUs and 40 NVIDIA RTX-600 Blackwell GPUs for inferencing to support NaviGator AI.
- To build the Digital Twin Hub, 360 RTX-6000 Blackwell GPUs are added to HiPerGator 4th generation.
- The replacement for the Orange storage system with 30 PB of capacity is acquired.
- "Building University Technological Infrastructure to Support AI" by Brittany Wise.

14K+

Students annually

UF enrollment in AI courses

268+

AI-related internships

Via the Career Connection Center

300+

AI-focused faculty

Throughout the university

"Clean data is the oxygen of AI." —Elias Eldayrie, Vice President and Chief Information Officer

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